02 · CHOOSE

9:30 – 9:40 · 10 minutes

Choose the Right Storage & Network

Pick the KPI that matters, then the system that delivers it

The decision framework that turns the overview into a choice — and sets up the hands-on lab that follows.

KPIS | WORKLOADS | FABRIC | PROFILING

KPI Key Performance Indicator

CHOOSE · THE SCORECARD

Primary and Secondary KPIs — know which one you're optimizing



THROUGHPUT

GB/s of bulk movement

DOMINATES FOR

big sequential training reads & checkpoint writes



LATENCY

time per single I/O (µs–ms)

DOMINATES FOR

anything on the critical path — incl. KV-cache reads



IOPS

operations per second

DOMINATES FOR

small-file / metadata storms (millions of opens)



TTFT

INFERENCE ERA

time to first token

DOMINATES FOR

inference serving — the user-visible one



CAPACITY

usable PB and $/TB

DOMINATES FOR

datasets, checkpoints, long retention



RELIABILITY

durability + availability

DOMINATES FOR

can you lose it, and is it there when needed

Secondary KPIs for Inference

- Inter-token latency (ITL / TPOT)
- End-to-end time to answer
- Tail ratio (p99 / mean) under load
- Throughput: tokens per second per GPU, goodput
- GPU utilization at the busy hour
- KV-cache hit rate, recomputed tokens avoided, GPU-seconds per task
- Recall at a latency budget; QPS at recall
- Index build time; end-to-end time to searchable
- Small-read capacity at a p99
- Time to first byte; listing; small-object
- Degraded-state p99
- Time to tenant; provable isolation
- Cost per usable PB

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TTFT Time To First Token **IOPS** Input/Output Operations Per Second **KPI** Key Performance Indicator **KV** Key-Value

The metrics AI teams watch — and what moves them

E2E = TTFT + ITL × (N − 1)

TTFT = tokenize + prefill + first decode
ITL (a.k.a. TPOT) = one decode step · **N** = output tokens



TTFT

prompt in → first token out; prefill-dominated

STORAGE LEVER

Model-weight load on cold start. A shared prefix/KV tier lets a cache hit skip prefill entirely.

ITL / TPOT

each decode step; what makes streaming feel smooth

STORAGE LEVER

KV read latency when context has spilled off-GPU — stripe, pool and DoM choices show up here.

E2E

the whole response; what an agent waits on

STORAGE LEVER

Agentic chains multiply it — every storage stall in the loop compounds.

TPS / RPS

tokens or requests per second; the cost lever

STORAGE LEVER

KV-tier capacity → more concurrent sessions per GPU; bigger, more uniform batches.

Measure like-for-like. TPS is easy to inflate — shorter inputs cut TTFT and flatter the number. Same workload, same lengths, relative deltas. *Metric definitions after Hands-On LLM Serving and Optimization, Ch. 4.*

AI Artificial Intelligence **DoM** Data-on-MDT **TTFT** Time To First Token **ITL** Inter-Token Latency **TPOT** Time Per Output Token **E2E** End-to-End latency **TPS** Tokens Per Second **RPS** Requests Per Second **KV** Key-Value **LLM** Large Language Model

CHOOSE · THE DECISION

Match the KPI to the workload

WORKLOAD	KPI THAT DOMINATES	STORAGE	NETWORK
Large sequential training	THROUGHPUT	Parallel FS (Lustre/EXAScaler)	RDMA / IB, GPUDirect
Checkpoint writes	THROUGHPUT + BURST	Parallel FS, wide stripe	High-bandwidth fabric
Inference / KV-cache serving	LATENCY + TTFT	NVMe tier / object KV-native	NVMe-oF, low-latency RDMA
Billions of small files	IOPS	Object, or DoM/DNE on parallel FS	Standard, scale-out
Archive / retention	CAPACITY ($/TB)	Object, tiered	Commodity Ethernet

“Pick the row you live in first. The storage and network columns fall out of it.”

ddn

DNE Distributed Namespace **DoM** Data-on-MDT **NVMe-oF** NVMe over Fabrics **NVMe** Non-Volatile Memory Express **RDMA** Remote Direct Memory Access **IB** InfiniBand **TTFT** Time To First Token **IOPS** Input/Output Operations Per Second **KPI** Key Performance Indicator **KV** Key-Value **FS** File System

CHOOSE · DON’T FORGET THE WIRE

Right network, then profile before you tune

Ethernet / IB
the baseline fabric

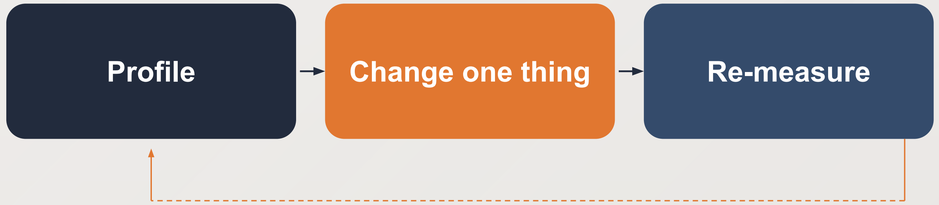
RDMA / RoCE
kernel-bypass, low latency

GPUDirect
NIC ↔ GPU, CPU out of the path

NVMe-oF
remote flash ≈ local flash

Even the fastest storage in the world waits when connected to the wrong network. Match the fabric to the KPI, not the brochure.

THEN: PROFILE BEFORE YOU TUNE



You cannot choose a KPI you haven’t measured. Every fix in today’s three labs starts from a profile:

LAB 1 — you profiled a single node in Nsight Systems and found the small-I/O stall.

LAB 2 — change the storage config, then measure the delta.

NVMe-oF NVMe over Fabrics **RDMA** Remote Direct Memory Access **RoCE** RDMA over Converged Ethernet **IB** InfiniBand **KPI** Key Performance Indicator